# The threat of synthetic respondents extends to clinical mental health screening

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## Key Points

**Question** Can commercially available large language models generate clinically plausible and severity-sensitive responses on validated psychiatric screening instruments?

**Findings** In this simulation study of 2106 synthetic personas, a standard commercial LLM produced diagnosis-congruent scores exceeding clinical cutoffs across all seven instruments, with scores scaling monotonically with assigned severity—without any instrument content or scoring information in the prompts.

**Meaning** Coherent symptom endorsement above clinical thresholds can no longer serve as a proxy for authentic participation in online mental health research, making the development of detection methods calibrated to clinical response patterns an urgent priority.

## Abstract

**Importance** Large language models (LLMs) can autonomously complete online surveys with human-like plausibility, evade standard quality-control checks, and do so at negligible cost. This poses a serious threat to the integrity of remotely collected research data. Whether this threat extends to validated clinical psychiatric screening instruments, which serve as gatekeepers for study eligibility and primary outcome measures in mental health research, has not yet been examined.

**Objective** To determine whether a commercially available LLM, provided only with brief diagnostic persona descriptions, can produce clinically differentiated and severity-sensitive responses across a broad battery of validated psychiatric screening instruments.

**Design, Setting, and Participants** This simulation study used Google Gemini 2.0 Flash to generate 2106 unique synthetic personas from 13 DSM-informed clinical diagnostic profiles, three severity levels, and demographic variables. Diagnostic profiles spanned mood disorders, anxiety disorders, obsessive-compulsive disorder, psychotic disorders, eating disorders, and neurodegenerative conditions.

**Main Outcomes and Measures** Scores on seven validated clinical instruments administered to each persona: the Patient Health Questionnaire–9 (PHQ-9), Generalized Anxiety Disorder–7 (GAD-7), Obsessive-Compulsive Inventory–Revised (OCI-R), PTSD Checklist for DSM-5 (PCL-5), Mood Disorder Questionnaire (MDQ), Eating Disorder Examination Questionnaire (EDE-Q), and Prodromal Questionnaire–16 (PQ-16). The primary outcome was whether synthetic personas generated diagnosis-congruent scores exceeding established clinical cutoffs.

**Results** LLM-generated personas produced clinically differentiated patterns of diagnostic specificity across groups and instruments. Five of seven instruments showed significantly higher scores among diagnosis-congruent personas (all $P < .001$), and the two

instruments typically elevated across multiple disorders (PHQ-9 and GAD-7) showed no differentiation between target and nontarget clinical controls ($P = .915$ and $P = .306$, respectively). Assigned severity modulated scores monotonically across all seven instruments (all $P < .001$).

**Conclusions and Relevance** A standard commercial LLM, given only brief diagnostic descriptions, can generate clinically plausible and severity-sensitive responses on validated psychiatric screening instruments without any specialized configuration. Coherent symptom endorsement above clinical thresholds can no longer serve as a proxy for authentic participation when recruiting online samples, making the development of detection methods calibrated to clinical response patterns an urgent priority.

Online platforms and crowdsourcing services have transformed how clinical researchers study mental health.[1] Surveys administered through services such as Amazon Mechanical Turk and Prolific enable rapid, low-cost access to large and diverse samples, including populations that are difficult to reach through traditional recruitment due to stigma, geographic barriers, or low base rates.[2] This infrastructure has become particularly important for research on psychiatric conditions, where validated screening instruments commonly serve as gatekeepers for study eligibility and primary outcome measures. However, the assumption that coherent, contextually appropriate survey responses require a human respondent is no longer tenable.

Westwood recently demonstrated that autonomous LLM-based respondents can complete online surveys with high fidelity.[3] These LLM-based agents passed attention checks, produced consistent psychometric profiles, and evaded standard detection methods. Critically, each completed survey cost approximately $0.005, making large-scale fabrication economically trivial. Moreover, the average user can now employ AI tools with agents built into browsers that can take surveys semi-autonomously with little input.[4] While the current scale of AI contamination on online research platforms remains uncertain, these initial probes demonstrate that the threat to online research is broad and deepening.[5] Yet none of these studies examined whether LLMs can produce plausible responses on the clinical instruments that mental health researchers regularly rely on to screen, characterize, and study psychiatric populations.

Clinical psychiatric instruments may be especially vulnerable to LLM mimicry for several reasons. First, mental health content, including descriptions of symptom presentations, diagnostic criteria, and self-report questionnaire items, is extensively represented in the text corpora used to train these models.[6] For example, word embeddings trained exclusively on LLM-generated synthetic self-reports can predict mental health screening scores with high accuracy,[7] suggesting that even synthetic clinical text may capture patterns associated with diagnostic information. Second, many clinical studies recruit

participants online precisely because the populations of interest are difficult to access through other means.[2] Third, researchers routinely rely on published cutoff scores to validate study eligibility, treating coherent symptom endorsement as implicit proof of a genuine respondent. An agent need only exceed a known threshold to infiltrate a clinical sample, turning quality-control metrics into a vulnerability.

# Methods

## Synthetic Persona Generation

We generated 2106 unique synthetic personas by taking the Cartesian product of 13 clinical diagnostic profiles, three severity levels (mild, moderate, severe), three age groups (25, 50, 75 years), two genders (male, female), three education levels (high school, bachelor's degree, master's degree), and three political affiliations (Democrat, Republican, Independent). Each clinical profile specified a DSM-informed diagnosis and characteristic symptoms. The 13 diagnoses spanned mood disorders (major depressive disorder, bipolar I disorder, alcohol-induced mood disorder), anxiety disorders (generalized anxiety disorder, posttraumatic stress disorder), obsessive-compulsive disorder, psychotic disorders (schizophrenia), eating disorders (anorexia nervosa, bulimia nervosa, binge eating disorder), and neurodegenerative conditions (Alzheimer disease, Parkinson disease dementia, frontotemporal dementia).

## Clinical Assessment Batteries

Each persona completed seven clinical instruments: (1) the Patient Health Questionnaire–9 (PHQ-9; nine items, 0–3 scale);[8] (2) the Generalized Anxiety Disorder–7 (GAD-7; seven items, 0–3 scale);[9] (3) the Obsessive-Compulsive Inventory–Revised (OCI-R; 18 items, 0–4 scale);[10] (4) the PTSD Checklist for DSM-5 (PCL-5; 20 items, 0–4 scale);[11] (5) the Mood Disorder Questionnaire (MDQ; 13 yes/no items);[12] (6) the Prodromal Questionnaire–16

(PQ-16; 16 true/false items);[13] and (7) the Eating Disorder Examination Questionnaire (EDE-Q 6.0; 28 items, mixed frequency and attitudinal scales).[14] All items and response options were implemented as specified in the published instruments.

## Simulation Procedure

Simulations were conducted using the Google Gemini 2.0 Flash API. For each persona, the model received a system prompt containing: (1) the persona profile (diagnosis, symptoms, severity, demographics); (2) all prior responses from previous batteries, ensuring cross-instrument consistency; (3) the current battery's questions with response options; and (4) instructions to respond in character, maintain consistency, and calibrate language to the persona's education level. No adversarial prompt engineering or fine-tuning was employed. The prompt contained no information about instrument content, scoring rubrics, or clinical cutoffs.

## Scoring and Statistical Analysis

Responses were mapped to scores using standard scoring rubrics for each instrument. We assessed diagnostic specificity using a linear regression of scores on a target indicator (diagnosis-congruent vs nontarget) per battery, with cluster-robust standard errors by diagnosis. Severity monotonicity was tested by regressing scores on ordinal severity within target groups.

# Results

LLM-generated personas produced clinically differentiated response patterns across all seven instruments (**Figure 1**). We note two key findings.

First, diagnostic specificity was battery-dependent (**Table 1**; **Figure 1**). On disorder-specific instruments, target personas scored substantially higher than nontarget

controls: OCI-R ($b = 15.76$, 95% CI [12.20, 19.31], $P < .001$), PCL-5 ($b = 14.63$, 95% CI [10.09, 19.17], $P < .001$), MDQ ($b = 9.24$, 95% CI [8.42, 10.07], $P < .001$), PQ-16 ($b = 6.15$, 95% CI [3.02, 9.28], $P < .001$), and EDE-Q ($b = 2.95$, 95% CI [2.26, 3.64], $P < .001$). By contrast, the PHQ-9 and GAD-7 showed no diagnostic specificity: target and nontarget personas produced nearly identical mean scores (PHQ-9: 13.55 vs 13.42, $P = .915$; GAD-7: 10.78 vs 11.54, $P = .306$). Although the model generated elevated depression and anxiety symptoms across all diagnostic profiles, this is consistent with their known comorbidities with other disorders.[15]

Second, despite this specificity gap, assigned severity modulated scores monotonically within target groups on all seven instruments (all $P < .001$; **Table 1**). On the PHQ-9, for example, target personas scored 6.9 at mild, 12.8 at moderate, and 21.0 at severe levels — crossing the clinical cutoff (≥10) between mild and moderate severity. Similarly, PTSD personas scored 7.3, 33.4, and 66.0 on the PCL-5, exceeding the clinical cutoff (≥33) at moderate severity and nearly doubling it at severe.

Table 1: Diagnostic Specificity and Severity Monotonicity Across Clinical Instruments

| | Diagnostic Specificity[a] | | | Severity Monotonicity[b] | | |
|---|---|---|---|---|---|---|
| Instrument | $b$ | 95% CI | $P$ Value | $b$ | 95% CI | $P$ Value |
| PHQ-9 | 0.13 | [−2.25, 2.51] | .915 | 7.06 | [6.18, 7.94] | <.001 |
| GAD-7 | −0.76 | [−2.21, 0.69] | .306 | 6.91 | [6.49, 7.33] | <.001 |
| OCI-R | 15.76 | [12.20, 19.31] | <.001 | 18.32 | [16.90, 19.75] | <.001 |
| PCL-5 | 14.63 | [10.09, 19.17] | <.001 | 29.33 | [28.29, 30.37] | <.001 |
| MDQ | 9.24 | [8.42, 10.07] | <.001 | 3.56 | [2.75, 4.38] | <.001 |
| PQ-16 | 6.15 | [3.02, 9.28] | <.001 | 4.84 | [3.97, 5.70] | <.001 |
| EDE-Q | 2.95 | [2.26, 3.64] | <.001 | 1.70 | [1.26, 2.15] | <.001 |

Abbreviations: EDE-Q, Eating Disorder Examination Questionnaire; GAD-7, Generalized Anxiety Disorder–7; MDQ, Mood Disorder Questionnaire; OCI-R, Obsessive-Compulsive Inventory–Revised; PCL-5, PTSD Checklist for DSM-5; PHQ-9, Patient Health Questionnaire–9; PQ-16, Prodromal Questionnaire–16.[a]Regression coefficient ($b$) for the target indicator (diagnosis-congruent = 1 vs nontarget = 0) predicting instrument score, with cluster-robust standard errors by diagnosis. [b]Regression coefficient ($b$) for ordinal severity (1 = mild, 2 = moderate, 3 = severe) predicting instrument score within target groups, with cluster-robust standard errors by diagnosis. $N = 2106$ synthetic personas. The dependent variable is the total instrument score (EDE-Q uses the global mean score, range 0–6). All $P$ values are two-sided.

# Discussion

The triviality of generating plausible counterfeits illustrates the severity of this threat to mental health research. Given that a standard commercial LLM can semi-autonomously produce clinically differentiated profiles with zero specialized configuration, the economic incentives for survey fraud are now sharply asymmetric: fabricating a large, threshold-crossing clinical cohort costs almost nothing relative to the high costs of genuine clinical data collection. Our findings do not demonstrate that such contamination is currently occurring at scale, nor do we claim that our simulation exercise captures the full complexity of authentic clinical presentations. Rather, we highlight that the technical barrier to generating plausible clinical data is remarkably low, and that the field's current defenses may not be designed to address this class of threat. The patterns of diagnostic specificity and severity sensitivity shown here were achieved with prompts that contained no information about instrument content, scoring rubrics, or clinical cutoffs. Researchers should consider implementing identity verification beyond attention checks, triangulating self-report with behavioral or physiological measures, avoiding diagnosis-specific language in recruitment materials, and treating unverified online panel data with caution.[16] Developing standardized detection methods calibrated to clinical response patterns is therefore an urgent priority.

# Conclusions

A standard commercial LLM, given only brief diagnostic descriptions and no instrument content or scoring information, can generate clinically plausible and severity-sensitive responses on validated psychiatric screening instruments. Coherent symptom endorsement above clinical thresholds can no longer serve as a reliable proxy for authentic human participation in online mental health research.

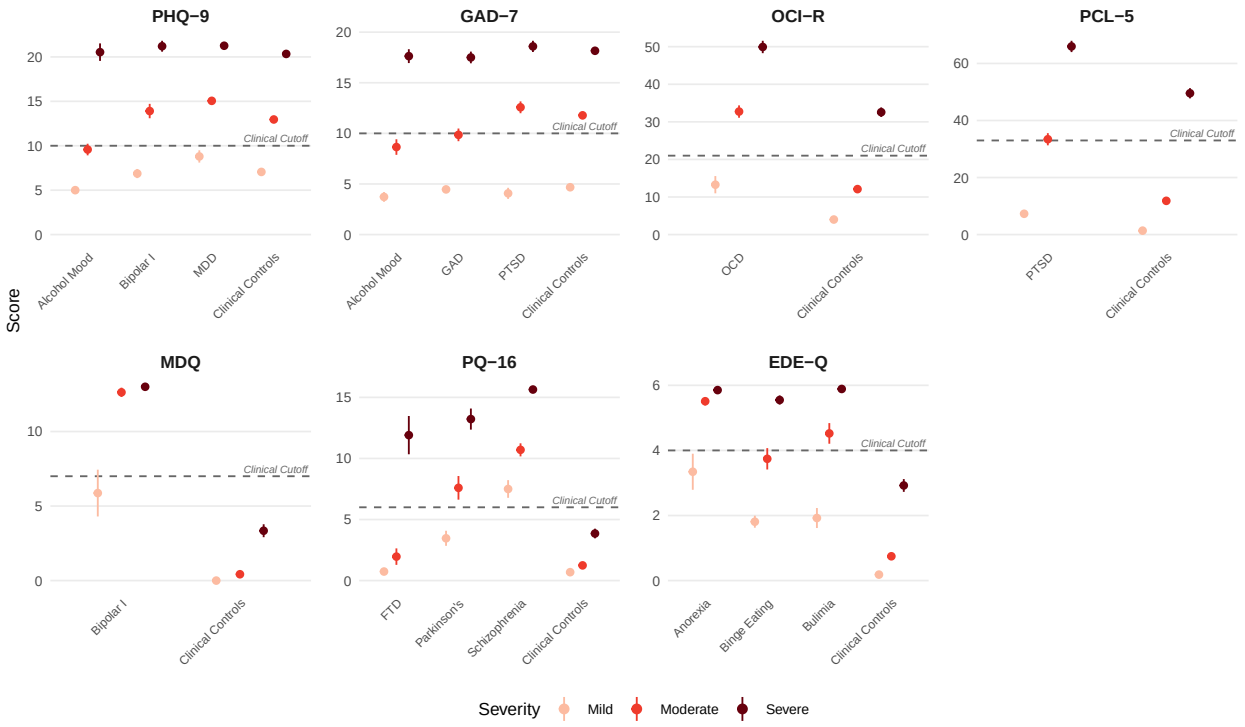


Figure 1: Synthetic respondent scores across seven validated clinical instruments. Each panel shows the mean score (95% CI) for clinically relevant diagnostic groups and an averaged clinical control group (all nontarget diagnoses) at three severity levels (light: mild; medium: moderate; dark: severe). Dashed lines indicate published clinical cutoff scores: PHQ-9 $\geq$ 10; GAD-7 $\geq$ 10; OCI-R $\geq$ 21; PCL-5 $\geq$ 33; MDQ $\geq$ 7; PQ-16 $\geq$ 6; EDE-Q $\geq$ 2.3. EDE-Q scores reflect the global mean (0–6 scale); all other instruments use summed totals. $n = 2106$ synthetic personas generated by Google Gemini 2.0 Flash.

# Acknowledgments

## Conflict of Interest Disclosures

LAB is a scientific advisor to Juniver, Ltd. The remaining authors report no conflicts of interest.

## Data and Code Availability

All simulation code, scoring scripts, and raw data are available at `https://github.com/kiante-fernandez/LLM-clinical`.

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